BRAIN TO BRAIN DECODER Neurodyads EEG

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Current year: Second year

>ELIGIBILITY AND AVAILABILITY:

I confirm that I meet the Google summer of code 2026 eligiblity requirements and that I am eligible to participate from India for the duration of the program. I am a full time undergraduate student, and my current status does not restrict me from participating in a paid open-source program.

During the GSOC period my fixed weekly commitments are mainly courseworks, selfstudy, and personal academic responsibilities, which together takes 28-35 hours per week. I can reliably dedicate about 25 to 35 focused hours each week to this 350 hour project, and I have a stable internet access for regular communication through emails, chat, and Github. If any examination-related disruption occurs, I will communicate with the mentors early and adjust the schedule while protecting the core milestones.

>Synopsis

This project aims to builds the repreodulible brain to brain decoding pipeline for the ML4SCI NeuroDyads, project using hyper scanning EEG data. Starting from the pre-task notebook I have already implemented, I will improve preprocessing, organize synchronised dyadic feature construction, train and elavuate CEBRA embeddings more systematically, and strengthen the scientific validity of the results through repeated control analyses. The final outcome would be a clean research respository with re-suable scripts, notebooks, plots, and documentation that can be directly used and extended but the ML4SCI mentors and future contributors

The NeuroDyads project focuses on learning low-dimentional neural embiddings from EEG hyper-scalling data collected from pairs of interesting participants. The ML4SCI project description emphasis the use of modern representation learning methods such as CEBRA to study whether dyadic EEG contains meaningful latent structure related to social interaction and task conditions.

The project is a strong fit for me because it sits at the intersection of machine learning, scientific Python, and EEG analysis, which are the areas I most want to develop further through open-source work. It is especially compelling because the project is not only about the training the model, but

about building a reliable pipeline when preprocessing, artifact handling, latent representations, controls and also the interpretation which basically matters.

As of for preparation I have completed the NeuroDyads pre-task notebook idependently. In that work, I loaded raw EDF files, segmented the EEG using the DIN1 markers which I have removed the VREF channel, which is applied ICA-based artificant correcvtion, generated power spectrum plots before and after ICA, created a joint 128-channel dyad representation, trained a 3D CEBRA embeddings, this is visualizing the embedding by task tables, and the implemented a shuffled-label control analysis with KNN evaluation. This gave me a pratical familiarity with the exact dataset, workflow as well as the challenges involved in the project.

>Benefits to the ml4sci community

The benefits to the ml4sci community is that it would transform an initial explanatory notebook into a structured and resuable open- source research pipeline. Instead of leaving the work at the level of a one-off experiment, this project will provide maintainable preprocessing scripts, cleaner embedding experiments, also this projects will also provides the maintainable preprocessing scripts, cleaner embedding experiments, repeatable controls and then the documentation which future contributors can build on directly

Also this project has enlighten me a concrete example which would provide how self-supervised or representation learning methods can be applied to hyper-scanning.EEG in an open-reproducible way. This helps more than one work as this could be beyond more than one projects as it demonstrated the workflow which can be scientific ml contributors can adapt to scientific neurosciences problem.

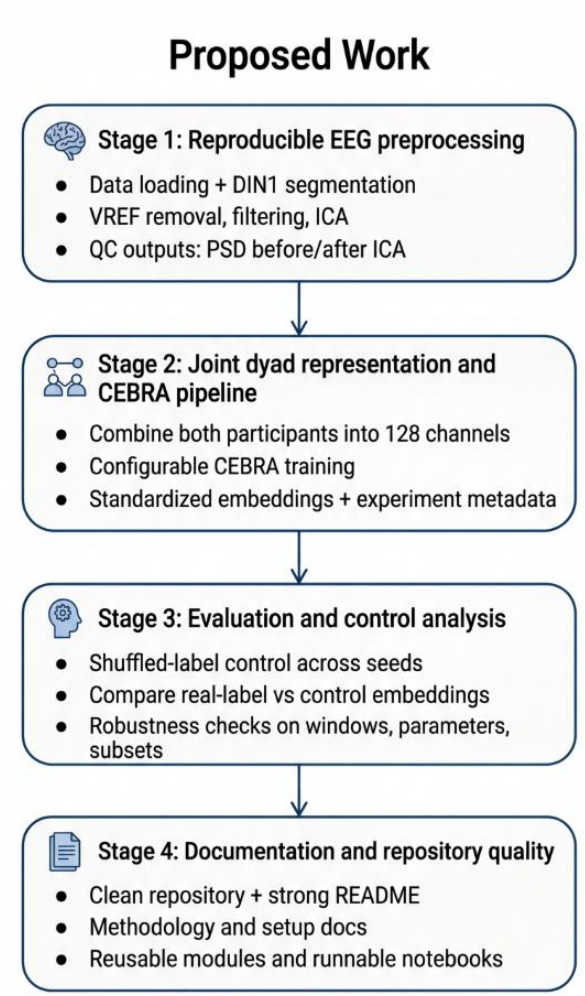
>Related work and current understanding

Before writing this proposal, I actually studied from the description provided in the project structure of the existing ml4sci NeuroDyads repository describes how the proposed work should fit the organisation direction. The repository also describes about how the toolkit is used for extracting and interpreting low-dimensional neural embeddings from the EEG hyperscanning of dyadic social interactions using the CEBRA framework, which matches the direction I have followed in pre-task implementation.

This pre-task in which I have worked on actually defines about how it helped me in several pratical issues that are easy to underestimate the project alone….so like I have worked on the segmentation around the event and was liked to be markers to be handled carefully and the ICA decisions it also affects the signal quality then we have the dyadic features matrix as well as the embedding quality , dyadic alignment matters to the work when constructing the feature matrix, and then the embedding quality must be judged during the controls analysis rathr than institution alone. Moreover these observations directs to motivate to work the plan proposed below.

>PROPOSED WORK

The project will be proceeded in the four connected stages



>DELIVERABLE

- A Resuable preprocessing module for neurodyads EEG data, Includes loading,DIN1-based segmentation, referencing, filtering, ICA, and then QC outputs.
- Clean notebooks which demonstrates the end to end pipelines from raw data to the embeddings and evaluation.
- Power spectrum plots before and then after ICA for representing participants and task segments
- A configurable CEBRA training pipeline for dyadic EEG embedding
- 3D embedding visualizations colored by label, with associated quantitative evaluation such as KNN accuracy in the embedding space
- Respective suffled-label control analysis and a clear comparison between true label and control results.
- Documentation covering setup, data placement methodology, output and limitation.

>Optional Deliverables

- Additional comparisons across embedding hyperparameters or label schemes.
- Limited robustness experiments across participants subset or different time windows.
- A light weight interactive visualisation script for exploring embedding.

>TECHNICAL APPROACH

Implementation will rely on Python scientific computing tools that are already part of my workflow. EEG preprocessing and event -based segmentation will use MNE and related libreries, while embedding and evaluation will use CEBRA, Numpy, pandas, mathplotlib, and scikit-learn style utilities. This ius also consistent with the tools I used in the Pre-task notebook.

>The high level Pipeline will be:

1.Load Paired EEG recordings,

2. Extract syncrhronized task windows from DIN1 markers.

3. Apply referencing, channel cleanup, filtering, and ICA.

4. Build the joint dyad representation.

5. Train CEBRA embedding using task labels.

6. Visualize the Embeddings in 3D.

7.Evaluate embeddings quantitatively.

8. Compare against shuffled-label controls.

9.document result and limitations clearly.

>TIMELINE

This is planned as realistic 350- hour project with slightly conservative scope, because completing a smaller, polished project is better than leaving as over ambitious one incomplete, that is also consistent with official GSoC guidance on proposal quality and feasibility.

Community bonding period

1. Rerun my current NeuroDyads Pre-task notebook from scratch.
2. Clarify mentor expectations and preferred repository structure.
3. Prepare a clean baseline branch and experiment log.

Week 1 to 2

Cleanup existing notebook so it runs end to end with minimal manual intervention.

Reproduce reprocessing and PSD outputs for selected dyads.

Record fragile steps and convert them into explicit functions.

Milestone: stable notebook with verified processing outputs.

Week 3 to 4

1. Refactor loading, segmentation, filtering, and ICA into reusable scripts or modules.
2. Add QC helper functions and clearer output organisation.
3. Test the preprocessing module across multiple examples. pipeline with QC plots.

Milestone: Reusable preprocessing pipeline with QC plots.

Week 5 to 6

1. Implement a configurable CEBRA pipeline for training dyadic embeddings.
2. Run a small grid of controlled experiments and save parameters, embeddings and evaluation metrics.
3. Generate standardized 3D visualizations.

Milestone: Working experiment pipeline for CEBRA embeddings

Week 7 to 8

1. Extend shuffled-label controls across multiple random seeds.
2. Compare true-label and shuffled-label embeddings visually and quantitatively.
3. Explore one robustnesss check if time permits.

Milestone: Completed control analysis notebook and summary.

Week 9 to 10

1. Improve documentation and clean up notebook for readability and reuse.
2. Write methodology and usage notes in the repository.
3. Incorporate mentor feedback on intermediate outputs.

Milestone: Documented repository ready for broader use.

Week 11to 12

1. Final polish, bug fixes, small requested extensions, and reports preparation.
2. Ensure reproducibility of key figured and results.
3. Leave buffet time for issues found during final review.

Final milestone: All required deliverables completed and handed over in a reusable form.

>Biographical information and Qualification

I am a second year B.Tech student in computer science and Engineering with specialization in AI and ML at Hindustan Institute of Technology and Science, Chennai. My primary technical tools are python, Jupyter, Numpy, pandas, Scipy, Scikit-learn, and PyTorch, and I am comfortable working through debugging- heavy scientific workflow.

I have also worked on academic and personal ML projects involving data preprocessing, model experimentation, and interpretation, including EEG-related and BCI-style work at the collage level. These experiences helped me become comfortable with noicy data, iterative experimentation, and documenting my reasoning through notebooks and reports.

Most importantly for this proposal, I already completed the neuroDyads Pre-task independently and implemented all of the required analysis components myself. That gives me a practical starting point and lowers the execution risk of the project significantly.

>Ethical use of AI tools

I am aware that organisation can distinguish between acceptable AI-assisted debugging and unacceptable AI-generated analysis. In my preparation for this project, AI tools were used only to help interpret specific error messages, recall syntax, and improve wording, The Preprocessing choices, Latent-space design, control analysis logic, and interpretation notebook directly.

>Risk and Mitigation

The largest technical risk is that EEG preprocessing choices may strongly affect downstream embeddings. I will mitigate this by keeping reprocessing modular, generating QC plots regularly, and documenting every important choice rather than silently adjusting parameters.

A second risk is over scoping the project. To avoid that , the proposal is intentionally centered on extending work I have already completed in the pre-task, with optional robustness experiments only after the core pipeline is stable. This approach aligns with GSoC guidelines that a smaller finished project is better than a large incomplete one.

>Why this proposal is a strong Fit.

This proposal is a strong fit because it is grounded in real prior work, aligns directly with the ML4SCI NeuroDyads idea, and is scoped around concrete deliverables rather than vague gaols. I am not applying to explore the project from zero I am proposing to turn an already working baseline into reproducible and mentor-usable open source research pipeline. That is the kind of result I would be proud to deliver for ML4SCI.